CALCULUS II - AY2024-25

Chapter 2. Extremums of Functions of Several Variables

Outline

1. Directional Derivatives
2. Maximum & Minimum Rate of Change
3. Optimization Problems
4. Double Integrals

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• Directional Derivatives

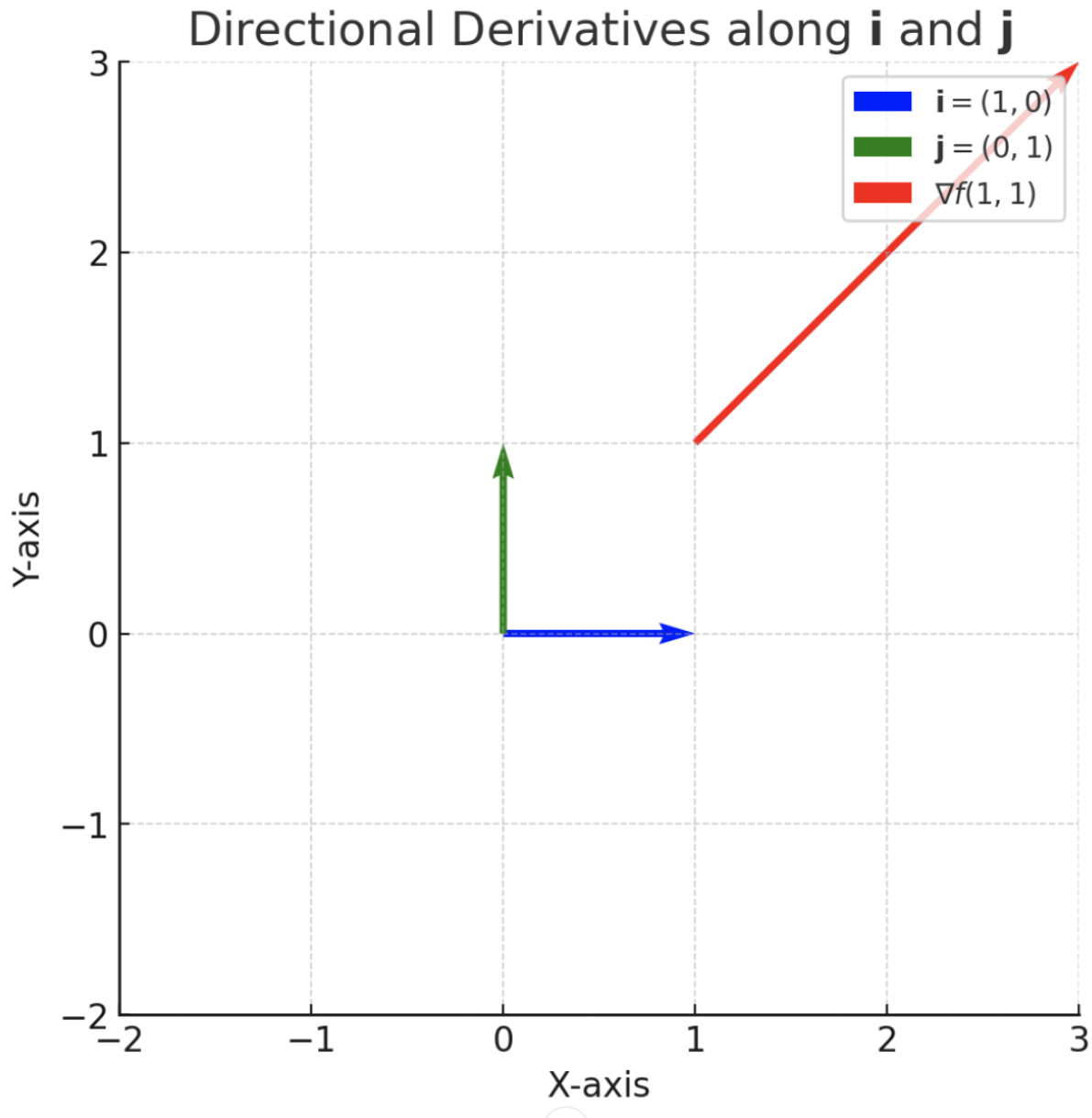
Recall. For the function $f(x, y)$, the partial derivatives f_x and f_y at a point (a, b) are defined as

$$f_x(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h},$$

and

$$f_y(a, b) = \lim_{h \to 0} \frac{f(a, b + h) - f(a, b)}{h}.$$

They represent respectively the rate of change of f in the x- and y-directions, i.e., the directions of the unit vectors **i** and **j**.


Directional Derivatives along i and j
i = (1, 0)
j = (0, 1)
∇f(1, 1)
Y-axis
X-axis
3
2
1
0
−1
−2
−2
−1
0
1
2
3

• Directional derivatives for 2-variable functions

These lead us to the following general notion.

Suppose that we want to find the rate of change of f in the direction corresponds to unit vector $\mathbf{u} = (u_1, u_2)$.

We follow the idea for functions of one variable.

- First note that the point changes from (a, b) to $(a, b) + h(u_1, u_2) = (a + hu_1, b + hu_2)$, and hence the corresponding change in f is

$$f(a + hu_1, b + hu_2) - f(a, b).$$

- Then the rate of change along the direction $\mathbf{u}$ is thus the limit of the ratio

$$\frac{f(a + hu_1, b + hu_2) - f(a, b)}{h} \quad \text{as } h \to 0.$$

We define this limit, if exists, as the directional derivative of f along $\mathbf{u}$.

Definition

The directional derivative of f at (a, b) in the direction of the unit vector $\mathbf{u} = (u_1, u_2)$ is

$$D_{\mathbf{u}} f(a, b) = \lim_{h \to 0} \frac{f(a + hu_1, b + hu_2) - f(a, b)}{h},$$

if this limit exists.

• Computation of $D_{\mathbf{u}}f$

The following result gives a formula for computing $D_{\mathbf{u}}f$.

Theorem

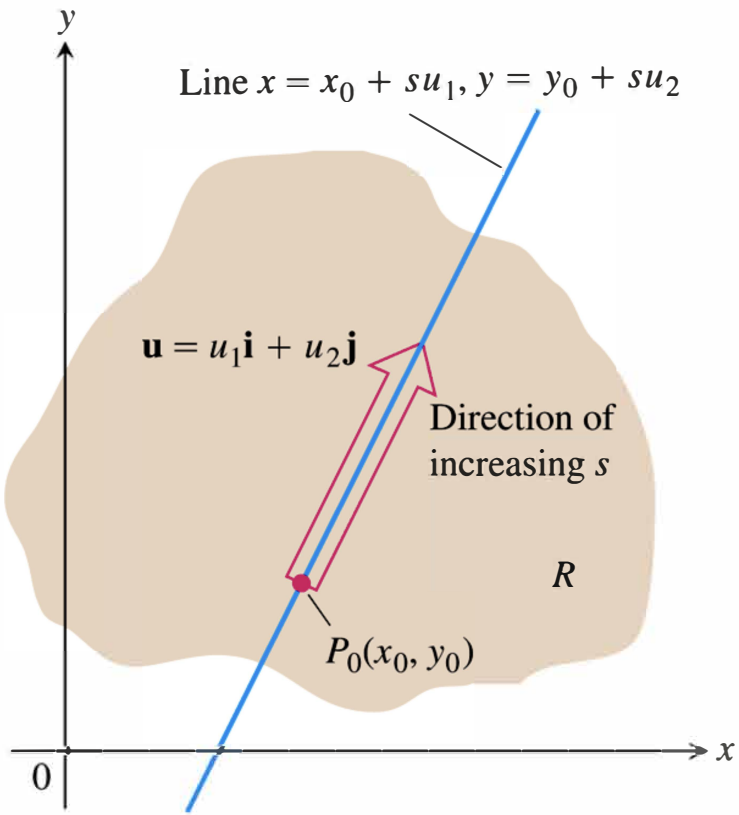
If $f(x, y)$ is a differentiable function of two variables x and y, then f has a directional derivative in the direction of any unit vector $\mathbf{u} = (u_1, u_2)$ and

$$D_{\mathbf{u}}f(x, y) = \nabla f \cdot \mathbf{u} = f_x(x, y)u_1 + f_y(x, y)u_2.$$

To evaluate $D_{\mathbf{u}}f(x, y)$ at $P(a, b)$ using the formula above, we compute

$$D_{\mathbf{u}}f(a, b) = \nabla f(a, b) \cdot \mathbf{u} = f_x(a, b)u_1 + f_y(a, b)u_2.$$

So we have to identify the point P and the (direction) unit vector $\mathbf{u}$.


y
Line $x = x_0 + su_1, y = y_0 + su_2$
$\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$
Direction of increasing s
R
$P_0(x_0, y_0)$
0
x

Example

Let $f(x,y) = x^3 - 3xy + 4y^2$.

(a) Find the directional derivative $D_{\mathbf{u}}f(x,y)$, where $\mathbf{u}$ is the unit vector given by angle $\theta = \dfrac{\pi}{6}$.

(b) What is $D_{\mathbf{u}}f(1,2)$?

Solution. We have $\mathbf{u} = \cos\frac{\pi}{6}\mathbf{i} + \sin\frac{\pi}{6}\mathbf{j} = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$, $P = (1,2)$, and $\nabla f = (3x^2 - 3y, -3x + 8y)$.

(a) $D_{\mathbf{u}}f(x,y) = \nabla f(x,y) \cdot \mathbf{u} = 3x^2 - 3y, -3x + 8y) \cdot \left(\dfrac{\sqrt{3}}{2}, \dfrac{1}{2}\right) = \dfrac{3\sqrt{3}}{2}x^2 - \dfrac{3}{2}x + 4y - \dfrac{3\sqrt{3}}{2}y.$

(b) $D_{\mathbf{u}}f(1,2) = \dfrac{3\sqrt{3}}{2} - \dfrac{3}{2} + 4 \cdot 2 - \dfrac{3\sqrt{3}}{2} \cdot 2 = \dfrac{13}{2} - \dfrac{3\sqrt{3}}{2}.$

Example

Find the directional derivative of the function $f(x, y) = x^2y^3 - 4y$ at the point $(2, -1)$, in the direction of the vector $\mathbf{v} = 2\mathbf{i} + 5\mathbf{j}$.

Solution. First we should find the unit vector **u** along **v**.

Since $\|\mathbf{v}\| = \sqrt{29}$, the unit vector in the direction of **v** is

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{1}{\sqrt{29}}(2, 5).$$

Also for $f(x, y) = x^2y^3 - 4y$, we have

$$\nabla f(x, y) = (2xy^3, 3x^2y^2 - 4).$$

Finally, $P = (2, -1)$.

Example (cont.)

Now we have

$$D_{\mathbf{u}} f(x,y) = \nabla f(x,y) \cdot \mathbf{u} = (2xy^3, 3x^2y^2 - 4) \cdot \frac{(2,5)}{\sqrt{29}},$$

and hence at the point $(2,-1)$, we obtain

$$D_{\mathbf{u}} f(2,-1) = (-4,8) \cdot \frac{(2,5)}{\sqrt{29}} = \frac{-4 \cdot 2 + 8 \cdot 5}{\sqrt{29}} = \frac{32}{\sqrt{29}}.$$

• Directional derivatives for 3-variable functions

For functions of three variables, directional derivatives are defined in the same manner.

Definition

The directional derivative of f at (a, b, c) in the direction of a unit vector $\mathbf{u} = (u_1, u_2, u_3)$ is

$$D_{\mathbf{u}} f(a, b, c) = \lim_{h \to 0} \frac{f(a + hu_1, b + hu_2, c + hu_3) - f(a, b, c)}{h},$$

if the limit exists.

We also have a formula

$$D_{\mathbf{u}} f(x, y, z) = \nabla f(x, y, z) \cdot \mathbf{u}.$$

• Directional derivatives for n-variable functions

Similarly, for a function of n variables $f(x_1, x_2, \ldots, x_n)$, we have the following definition.

Definition

The directional derivatives of f at $\mathbf{x_0}$ along the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}} f(\mathbf{x}_0) = \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{u}) - f(\mathbf{x}_0)}{h}.$$

The formula for computing a directional derivative is

$$D_{\mathbf{u}} f(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{u},$$

i.e.,

$$D_{\mathbf{u}} f(x_1, x_2, \ldots, x_n) = \nabla f(x_1, x_2, \ldots, x_n) \cdot \mathbf{u}.$$

Outline

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• Maximum & Minimum Rate of Change

We know that the value $D_{\mathbf{u}} f$ gives the rate of change in f along the direction $\mathbf{u}$.

Concerning this, there are two questions we seek answers to:

1. Does there exists a direction where the rate of change in f is the maximum or a minimum?
2. How to determine this maximum and minimum?

To give the answers, we look at the formula

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} = \|\nabla f\| \cdot \|\mathbf{u}\| \cos\theta = \|\nabla f\| \cos\theta,$$

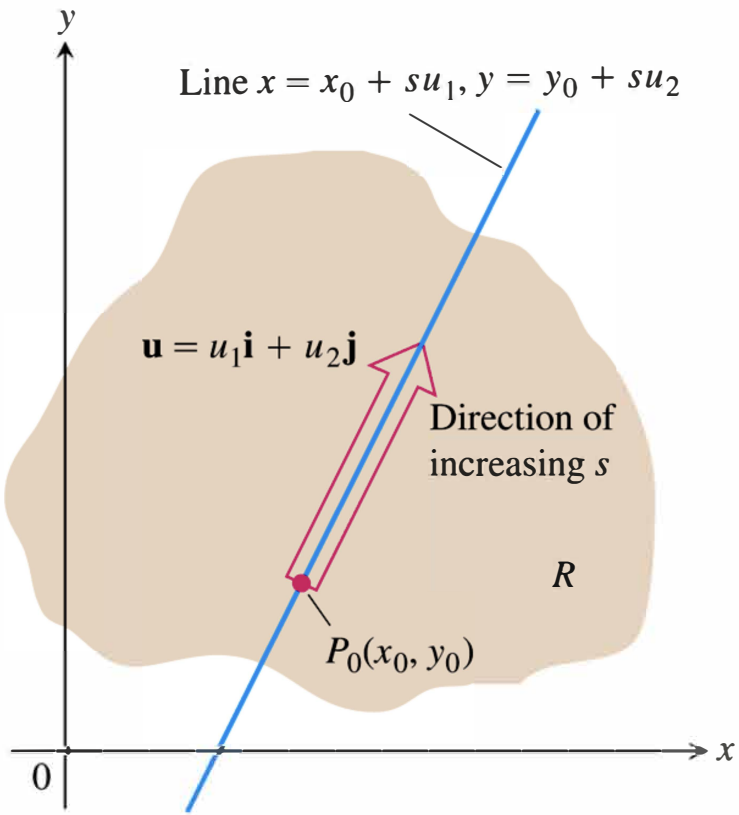
where θ is the angle between ∇f and $\mathbf{u}$.

Since $-1 \leq \cos\theta \leq 1$ ($\cos 0 = 1$ and $\cos\pi = -1$), we have

$$-\|\nabla f\| \leq \|\nabla f\| \cos\theta \leq \|\nabla f\|.$$

Equalities occur when $\cos\theta = -1 \iff \theta = \pi$ for the left inequality and $\cos\theta = 1 \iff \theta = 0$ for the right inequality.

Note that $\theta = 0$ means $\mathbf{u}$ has the same direction as the gradient vector ∇f and $\theta = \pi$ means $\mathbf{u}$ has the opposite direction of the gradient vector ∇f.



The rate of change of f in the direction of $\mathbf{u}$ at a point P_0 is the rate at which f changes along this line at P_0.

• Maximum & Minimum Rate of Change

So we obtain the following result.

Theorem

Suppose f is a differentiable function (of two or three variables).

(a) *The maximum rate of change in f at P_0 is $\|\nabla f(P_0)\|$ and it occurs in the direction $\mathbf{u} = \dfrac{\nabla f(P_0)}{\|\nabla f(P_0)\|}$, which is the same direction as the gradient vector $\nabla f(P_0)$.*

(b) *The minimum rate of change in f at P_0 is $-\|\nabla f(P_0)\|$ and it occurs in the direction $\mathbf{u} = \dfrac{-\nabla f(P_0)}{\|\nabla f(P_0)\|}$, which is the opposite direction of the gradient vector $\nabla f(P_0)$.*

Example

(a) For $f(x,y) = xe^y$, find the rate of change of f at the point $P(2,0)$ in the direction from P to $Q\left(\frac{1}{2}, 2\right)$.

(b) What is the maximum rate of change? In what direction does f have the maximum rate of change?

Solution.

(a) The rate of change of f at point P along a direction of the unit vector $\mathbf{u}$ is given by $D_{\mathbf{u}} f(P) = \nabla f(P) \cdot \mathbf{u}$.

- First, $\nabla f(x,y) = (e^y, xe^y) \implies \nabla f(2,0) = (e^0, 2e^0) = (1,2)$.

- Next, $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \left(-\frac{3}{2}, 2\right)$, and hence the unit vector in the direction $\overrightarrow{PQ}$ is $\mathbf{u} = \frac{\overrightarrow{PQ}}{\|\overrightarrow{PQ}\|} = \frac{\left(-\frac{3}{2},2\right)}{\sqrt{\left(-\frac{3}{2}\right)^2 + 2^2}} = \left(-\frac{3}{5}, \frac{4}{5}\right)$.

Then the rate of change of f in the direction from P to Q is

$$\begin{aligned} D_{\mathbf{u}} f(2,0) &= \nabla f(2,0) \cdot \mathbf{u} \\ &= (1,2) \cdot \left(-\frac{3}{5}, \frac{4}{5} \right) = -\frac{3}{5} + 2 \cdot \frac{4}{5} = 1. \end{aligned}$$

(b) We have computed $\nabla f(2,0) = (1,2)$. Then the maximum rate of change in f at P is

$$\|\nabla f(2,0)\| = \|(1,2)\| = \sqrt{1^2 + 2^2} = \sqrt{5}.$$

At P, the function f increases fastest in the direction of the gradient vector $\nabla f(2,0) = (1,2)$. The corresponding unit vector $\mathbf{u}$ is

$$\mathbf{u} = \frac{\nabla f(2,0)}{\|\nabla f(2,0)\|} = \frac{1}{\sqrt{5}}(1,2).$$

• Important Conclusion

1. The gradient vector ∇f is orthogonal to the level curve $f(x, y) = k$ and it is the direction where f increases most rapidly (largest rate).
2. The gradient vector ∇F is orthogonal to the level surface $F(x, y, z) = k$ and it is the direction where F increases most rapidly (largest rate).

Remark. The direction $-\nabla f(P)$ (respectively $-\nabla F(P)$ is used in the steepest descent algorithm in seeking the global minimum value of f (respectively of F).

• Singular, stationary, and critical points

Definition

(a) Points at which f_x or f_y do not exist are called singular points.

(b) If (a, b) is not a singular point, and $\nabla f(a, b) = (0, 0)$, i.e. both $f_x(a, b) = f_y(a, b) = 0$, then we call the point (a, b) a stationary point.

(c) A point (a, b) is called a critical point of f if (a, b) is a stationary point or a singular point, that is

$$\{\text{critical points}\} = \{\text{singular points}\} \bigcup \{\text{stationary points}\}.$$

Example

Find all critical points for $f(x,y) = y^2 - y^4 - x^2$.

Solution. First we note that

$$\nabla f(x,y) = (-2x, 2y - 4y^3) = \big(-2x, 2y(1-2y^2)\big),$$

is defined at every (x,y), and hence there is no singular point.

Next, to find stationary points, we solve the equation

$$\nabla f(x,y) = (0,0) \iff \begin{cases} -2x = 0 \\ 2y(1-2y^2) = 0 \end{cases} \iff \begin{cases} x = 0 \\ y = 0, \pm\dfrac{1}{\sqrt{2}}. \end{cases}$$

Thus stationary points are $(0,0)$, $(0, \frac{1}{\sqrt{2}})$, and $(0, \frac{-1}{\sqrt{2}})$ (which are all critical points of f).

Example

Find all critical points of $f(x,y) = e^{x\sqrt[3]{y}-x^2}$.

Solution. We have

$$\nabla f(x,y) = e^{x\sqrt[3]{y}-x^2}\left(\sqrt[3]{y} - 2x, \frac{x}{3\sqrt[3]{y^2}}\right), \quad \text{if } y \neq 0.$$

- First, $\nabla f(x,y)$ is not defined when $y = 0$, and hence singular points are $(x,0) : x \in \mathbb{R}$.

- Next, for stationary points, we solve $\nabla f(x,y) = \mathbf{0}$, with $y \neq 0$.

Example (cont.)

We have

$$\nabla f(x,y) = \mathbf{0} \iff \begin{cases} \sqrt[3]{y} - 2x = 0 \\ \dfrac{x}{3\sqrt[3]{y^2}} = 0. \end{cases}$$

From $\dfrac{x}{3\sqrt[3]{y^2}} = 0$, we get $x = 0$.

Substituting $x = 0$ into $\sqrt[3]{y} - 2x = 0$, we obtain $y = 0$.

But $y \neq 0$. So there is no stationary point.

Conclusion: Critical points of f are $(x, 0) : x \in \mathbb{R}$.

• Local maximum and local minimum values

In Calculus I we have discussed that one of the main applications of derivatives is to classify local extrema of single variable functions via first-order or second-order derivatives.

Now we see how to use partial derivatives to find and classify the local extrema of functions of two variables.

Definition

- A function f of two variables has a local maximum at (a, b) if

$$f(x, y) \leq f(a, b),$$

for all points (x, y) in some disk centered at (a, b).

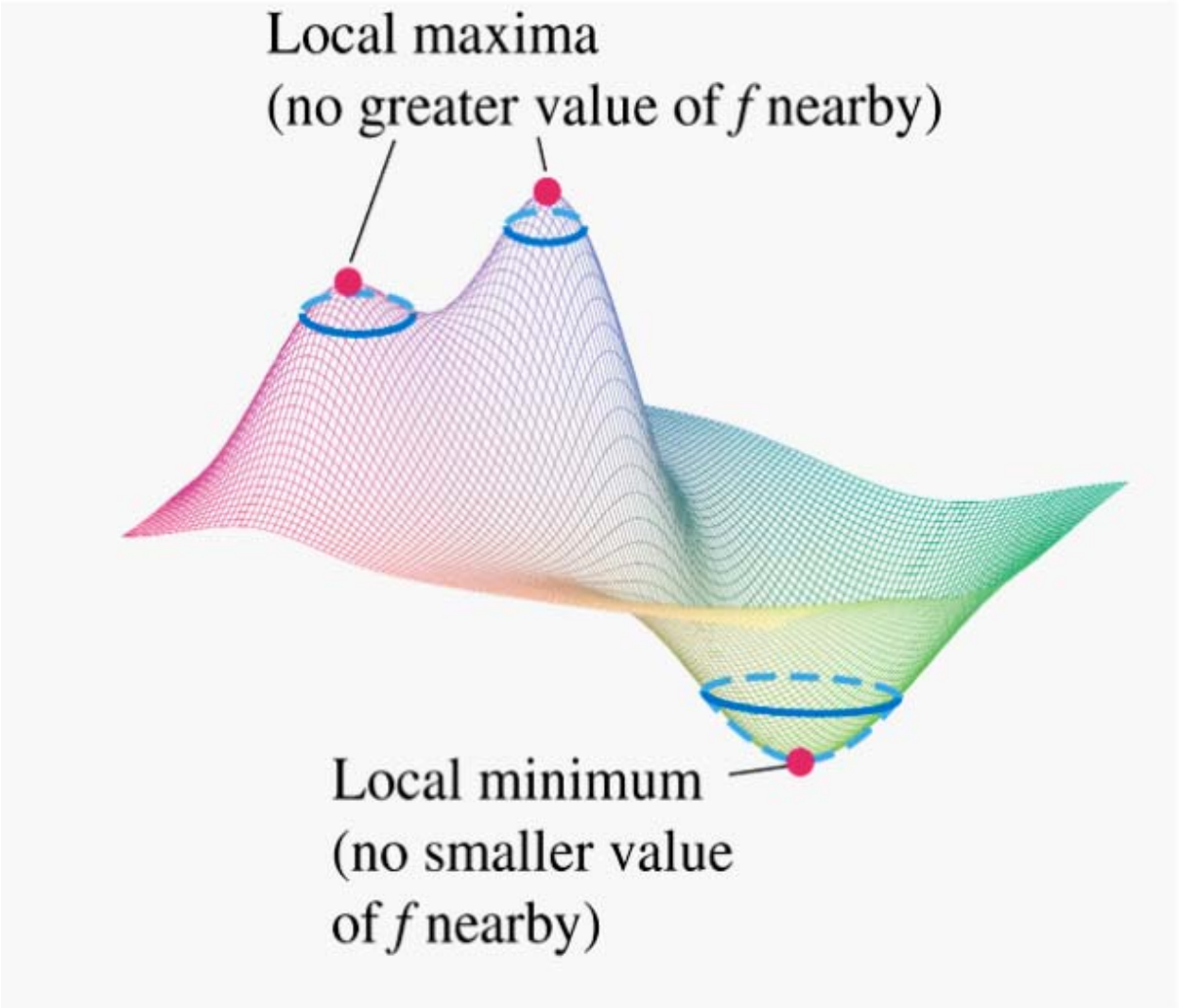
The number $f(a, b)$ is called a local maximum value.

- If the inequality above is changed to

$$f(x, y) \geq f(a, b),$$

then f has a local minimum at (a, b), and $f(a, b)$ is called a local minimum value.

• Local Extreme Values



• Necessary condition for local extreme values

The following result tells us possible points for local extrema are stationary points if both f_x and f_y exist.

Theorem

If f has a local extremum at (a, b), where (a, b) is an interior point, and the first-order partial derivatives exist there, then (a, b) is a stationary point, i.e.

$$\nabla f(a, b) = (0, 0).$$

• Saddle point

However, note that at a stationary point (a, b), the value $f(a, b)$ may or may not be a local maximum or a local minimum.

So we have the following definition.

Definition

A point (a, b) is called a saddle point of f if it is a stationary point but neither local maximum nor local minimum.

In other words, $f_x(a, b) = f_y(a, b) = 0$, but $f(a, b)$ is neither a local maximum nor local minimum.

Example (9. Back to one of examples)

Let $f(x,y) = y^2 - y^4 - x^2$.

(a) Explain why f has a local maximum at stationary points $(0, \pm\frac{1}{\sqrt{2}})$.

(b) Is $f(x,y)$ a local extremum at the stationary point $(0,0)$?

Solution. We have found above that stationary points of f are $(0,0)$, $(0, \frac{1}{\sqrt{2}})$, and $(0, -\frac{1}{\sqrt{2}})$.

(a) We rewrite $f(x,y)$ in completing squares as follows

$$f(x,y) = \frac{1}{4} - \left(y^2 - \frac{1}{2}\right)^2 - x^2 \leq \frac{1}{4}, \quad \text{for all } (x,y).$$

Also we have $f\left(0, \pm\frac{1}{\sqrt{2}}\right) = \frac{1}{4}$.

Thus both points $\left(0, \pm\frac{1}{\sqrt{2}}\right)$ give local maximum values. (In fact, they are global maximum.)

(b) The origin $(0,0)$ is a stationary point, and $f(0,0) = 0$.

Note that

- along $x = 0$, $f(x,y) = y^2 - y^4 = y^2(1-y^2) \geq 0 = f(0,0)$, if $-1 \leq y \leq 1$.

- along $y = 0$, $f(x,y) = -x^2 \leq 0 = f(0,0)$, for all x.

Thus every disk centered at $(0,0)$ contains points where f takes *positive values* as well as points where f is *negative*.

These show that value $f(0,0)$ is neither a local maximum nor a local minimum.

So the function $f(x,y) = y^2 - y^4 - x^2$ has a saddle point at $(0,0)$.

- Diagram: Saddle point of $f = y^2 - y^4 - x^2$

